

SR P-Quiz (Statistical Reasoning)

- **Quiz Type:** Practice Quiz
 - **Total Points:** 36
 - **Questions:** 36
 - **Due Date:** None
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Question 1 (1 pt)

A researcher collects a sample, computes summary statistics, and directly reports conclusions about the population without considering sampling variability. Which step of the statistical process has been most critically neglected? ☐ A. Probability ☐ B. Producing data ☐ C. Exploratory Data Analysis ☒ **D. Inference**

Question 2 (1 pt)

A study aims to estimate the average salary of all teachers in a city but collects data only from teachers in private schools. Which statement BEST describes the issue? ☐ A. The study lacks exploratory data analysis ☐ B. The sample is too small to perform inference ☒ C. The sample may not represent the population ☐ D. The population is incorrectly defined

Question 3 (1 pt)

Which of the following sequences correctly reflects the logic of the statistical process? ☐ A. Inference → Probability → Data Collection → EDA ☐ B. Probability → Data Collection → Inference → EDA ☐ C. EDA → Inference → Probability → Data

Question 4 (1 pt)

A poll reports that 65% of a sample supports a policy and concludes that exactly 65% of the population supports it. What is the BEST critique? ☐ A. The sample size is too large ☒ B. The conclusion ignores sampling variability ☐ C. The data are categorical ☐ D. The result should not be summarized

Question 5 (1 pt)

Which of the following BEST distinguishes Exploratory Data Analysis (EDA) from Inference? ☐ A. EDA uses graphs; inference does not ☒ B. EDA summarizes sample data, inference generalizes to population ☐ C. EDA applies only to categorical variables ☐ D. Inference is performed before EDA

Question 6 (1 pt)

Two datasets have identical medians and identical IQRs. However, Dataset A has several extreme values while Dataset B does not. Which statement is MOST accurate? ☐ A. Dataset A must have a larger IQR ☐ B. Dataset B must have a larger standard deviation ☒ C. Dataset A is likely to have a larger standard deviation ☐ D. Both datasets must have identical standard deviation

Question 7 (1 pt)

A dataset has: Median = 50, Q1 = 40, Q3 = 60, and two extreme values at 120 and 130.

Which statement is MOST appropriate? ☒ A. The outliers will not affect the IQR but will affect the range ☐ B. The IQR will increase significantly due to the extreme values ☐ C. The box in the boxplot will extend to 130 ☐ D. The median will shift toward the outliers

Question 8 (1 pt)

A researcher is analyzing income data, which is highly right-skewed. Which combination is MOST appropriate for summarizing this dataset? ☐ A. Mean and standard deviation ☐ B. Mean and range ☐ C. Mode and variance ☒ D. Median and IQR

Question 9 (1 pt)

A dataset has 9 ordered values. When computing Q1 and Q3, which of the following is CORRECT? ☐ A. The median is included in both halves ☒ B. The median is excluded from both halves ☐ C. Q1 and Q3 are always equal ☐ D. IQR cannot be computed for odd sample sizes

Question 10 (1 pt)

Two histograms show exam scores: Both have the same mean; Histogram A is tightly clustered; Histogram B is widely spread. Which statement is MOST accurate? ☐ A. Both have equal variability ☒ B. Histogram B has larger standard deviation ☐ C. Histogram A has larger range ☐ D. Means must differ due to spread

Question 11 (1 pt)

A scatterplot of hours studied (X) vs exam score (Y) shows a clear upward linear trend and several points far below the trend line. Which of the following is the MOST appropriate interpretation? ☐ A. The relationship is weak because of the outliers ☐ B. There is no relationship due to variability ☐ C. The form is non-linear ☒ D. The relationship is strong but contains influential outliers

Question 12 (1 pt)

A study examines the relationship between Type of diet (vegetarian, non-vegetarian) and Blood cholesterol level (mg/dL). Which is the BEST classification? ☒ A. C $\rightarrow$ Q (Categorical explanatory to Quantitative response) ☐ B. Q $\rightarrow$ Q ☐ C. Q $\rightarrow$ C ☐ D. C $\rightarrow$ C

Question 13 (1 pt)

A scatterplot shows a clear curved pattern where Y increases with X at first, then decreases. Which statement is MOST accurate? ☐ A. The relationship is negative ☐ B. The relationship has no direction ☒ C. The relationship is strong but non-linear ☐ D. The relationship is weak because it is not linear

Question 14 (1 pt)

A study finds that cities with more firefighters tend to have more fire damage. Which is the MOST plausible explanation? ☒ A. A lurking variable (size of fire) influences both ☐ B. Firefighters cause fire damage ☐ C. Fire damage causes more firefighters ☐ D. The relationship must be ignored

Question 15 (1 pt)

Two scatterplots show positive linear relationships: Plot A (points closely clustered around a line), Plot B (points widely scattered around a line). Which statement is MOST accurate? ☐ A. Plot B has stronger relationship due to more variability ☒ B. Plot A has stronger relationship due to tighter clustering ☐ C. Both have equal strength since both are linear ☐ D. Plot B has no relationship

Question 16 (1 pt)

A university wants to study all undergraduate students' stress levels, but selects a sample using a list of students enrolled in evening classes only. Which issue BEST describes this situation? ☐ A. Nonresponse bias ☐ B. Voluntary response bias ☒ C. Sampling frame bias ☐ D. Measurement bias

Question 17 (1 pt)

A university posts an online survey asking, "How satisfied are you with campus facilities?" Students choose whether or not to respond. Which conclusion is MOST appropriate? ☐ A. The sample is random if enough students respond ☒ B. The results are likely biased due to voluntary response ☐ C. The sample is representative if responses are anonymous ☐ D. Bias is eliminated if sample size is large

Question 18 (1 pt)

A researcher randomly selects 10 schools, then surveys all students in those schools. Which sampling method is used? ☐ A. Simple random sampling ☐ B. Stratified sampling ☐ C. Systematic sampling ☒ D. Cluster sampling

Question 19 (1 pt)

A study examines the effect of sleep on academic performance by recording students' natural sleep habits and measuring their grades. Which statement is MOST accurate? ☐ A. This is an experiment because sleep is measured ☐ B. This is an experiment because grades are outcomes ☐ C. This is neither observational nor experimental ☒ D. This is an observational study because no intervention occurs

Question 20 (1 pt)

A student claims, "The sampling distribution is just the distribution of the data in one sample." Which is the BEST evaluation? ☒ A. Incorrect, because it describes variability across many samples ☐ B. Correct, because both describe variability ☐ C. Correct, if the sample is large ☐ D. Incorrect, because it only applies to categorical data

Question 21 (1 pt)

As sample size increases, what happens to the sampling distribution of the sample mean? ☐ A. It becomes more spread out ☐ B. It becomes less centered ☒ C. It becomes more concentrated around the population mean ☐ D. It shifts away from the population mean

Question 22 (1 pt)

Which statement BEST describes the relationship between the sample mean and population mean? ☐ A. The sample mean always equals the population mean ☐ B.

The sample mean tends to overestimate the population mean ☒ C. The sample mean is an unbiased estimator of the population mean ☐ D. The sample mean is unrelated to the population mean

Question 23 (1 pt)

A population is highly skewed. Which statement is MOST accurate about the sampling distribution of the sample mean? ☐ A. It will always be skewed regardless of sample size ☒ B. It will become approximately normal for large sample sizes (Central Limit Theorem) ☐ C. It will match the population distribution exactly ☐ D. It cannot be determined

Question 24 (1 pt)

A dataset has: $Q1 = 20$, Median = 35, $Q3 = 50$. Which of the following values would be classified as an outlier using the $1.5 \times IQR$ rule? ($IQR = 50 - 20 = 30$. *Upper cutoff* = $50 + 1.5(30) = 95$. *Values > 95 are outliers*). ☐ A. 70 ☐ B. 80 ☐ C. 95 ☒ D. 110

Question 25 (1 pt)

A population has standard deviation $\sigma = 20$. Two samples are taken: Sample A ($n = 25$), Sample B ($n = 100$). What is the ratio of the standard error of Sample A to Sample B? ($SE_A = 20/\sqrt{25} = 4$; $SE_B = 20/\sqrt{100} = 2$; *Ratio* = $4/2 = 2:1$). ☒ A. 2 : 1 ☐ B. 1 : 1 ☐ C. 1 : 2 ☐ D. 4 : 1

Question 26 (1 pt)

What type of variable is Annual_income? [☒] A. Quantitative [☐] B. Categorical

Question 27 (1 pt)

Thirty-two students were asked the number of servings of fruits and vegetables they eat daily. The results are displayed in the histogram below. How many of the students surveyed eat at least 4 servings of fruits and vegetables daily? [☐] A. 4 [☐] B. 8 [☒] C. 12 [☐] D. 20

Question 28 (1 pt)

(Based on the question above) What percentage of the students surveyed eat no more than 3 servings of fruits and vegetables daily? (*20 out of 32 = 62.5%*) [☐] A. 20 [☐] B. 31.2% [☐] C. 37.5% [☒] D. 62.5%

Question 29 (1 pt)

What is the most likely shape of the distribution of age of death from trauma (accident, murder, suicide, drug overdose, etc.) when represented by a histogram? [☐] A. Symmetric – uniform [☐] B. Skewed left [☒] C. Skewed right [☐] D. Symmetric – unimodal

Question 30 (1 pt)

A recent survey asked 90 students, "How many hours do you spend on the computer in a typical day?" Of the 90 respondents: 3 said 1h, 5 said 2h, 15 said 3h, 25 said 4h, 20

said 5h, 15 said 6h, 5 said 7h, 1 said 8h, 1 said 9h. What is the average (mean) number of hours spent on the computer? (*Total hours = $3(1)+5(2)+15(3)+25(4)+20(5)+15(6)+5(7)+1(8)+1(9) = 400 / 90 \approx 4.44$*) ☐ A. 0.5 ☐ B. 4 ☒ C. 4.44 ☐ D. 5

Question 31 (1 pt)

Here are the number of hours that 9 students spend on the computer on a typical day: 1, 5, 5, 6, 7, 8, 11, 12, 15. What is the median number of hours spent on the computer? (*Ordered 5th item = 7*) ☐ A. 5 ☐ B. 6.5 ☒ C. 7 ☐ D. 7.5

Question 32 (1 pt)

A survey taken in a large statistics class contained the question: "What's the fastest you have driven a car (in miles per hour)?" The five-number summary for the 87 males surveyed is: min = 55, Q1 = 95, Median = 110, Q3 = 120, Max = 155. Should the largest observation in this data set be classified as an outlier? (*IQR = $120 - 95 = 25$. Upper fence = $120 + 1.5(25) = 157.5$. Since $155 < 157.5$, it is not an outlier.*) ☐ A. Yes ☒ B. No

Question 33 (1 pt)

The boxplot below displays ratings for TV shows during sweeps week: 50% of the shows have a rating greater than which value? (*Value corresponding to Median*) ☐ A. 11 ☒ B. 15.5 ☐ C. 14 ☐ D. 17

Question 34 (1 pt)

The Web site kiplinger.com published the “Best Values in Public Colleges”... What percentage of the graduates will have a debt greater than \$25,000 (corresponding to Q3)? ☒ A. 25% ☐ B. 50% ☐ C. 75% ☐ D. 81%

Question 35 (1 pt)

(Based on the question before) 50% of the debts owed are smaller than what amount (Median)? ☐ A. \$18,000 ☒ B. \$20,600 ☐ C. \$22,000 ☐ D. \$25,000

Question 36 (1 pt)

(Based on the question before) Within which interval of debts would you expect to find the largest number of graduates? ☐ A. 13,000–17,000 ☒ B. 17,000–21,000 ☐ C. 21,000–25,000 ☐ D. 25,000–29,000